

English	(1015CMD303361230018)		Test Pattern
		CLASSROOM CONTACT PROGRAMME (Academic Session : 2023-2024)	NEET (UG)
			MINOR
			21-05-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE : MEF-I (B)

Test Booklet Code

This Booklet contains 28 pages.

E3

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

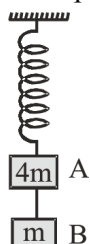
Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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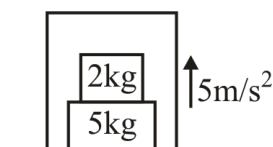
SECTION-A (PHYSICS)

- A body of mass m is taken to the bottom of a deep mine. Then :-
 (1) Its mass increases (2) Its mass decreases
 (3) Its mass is zero (4) Mass remain same
- If the radius of earth decreases by 1% but its mass remains constant then the value of acceleration due to gravity on earth surface will be :-
 (1) increases by 2% (2) decreases by 2%
 (3) increases by 1% (4) decreases by 1%
- What is the ratio of potential energy to kinetic energy of the moon orbiting around the earth?
 (1) 1 : 4 (2) 1 : 2 (3) 4 : 1 (4) 2 : 1
- If three particles each of mass M are placed at the corners of an equilateral triangle of side a , the potential energy of the system and the work done if the side of the triangle is changed from a to $2a$, are-
 (1) $\frac{3GM}{a}, \frac{3GM}{2a}$ (2) $-\frac{3GM^2}{a}, \frac{3GM^2}{2a}$
 (3) $-\frac{3GM^2}{a}, \frac{3GM^2}{4a^2}$ (4) $-\frac{3GM^2}{a}, \frac{3GM}{2a}$
- Two planets of radii in the ratio 2 : 3 are made from the materials of density in the ratio 3 : 2. Then the ratio of acceleration due to gravity $\frac{g_1}{g_2}$ at the surface of two planets will be :-
 (1) 1 (2) 2.25 (3) $\frac{4}{9}$ (4) 0.12
- Two blocks A and B of masses 4 m and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A & B immediately after the string is cut, are respectively:-

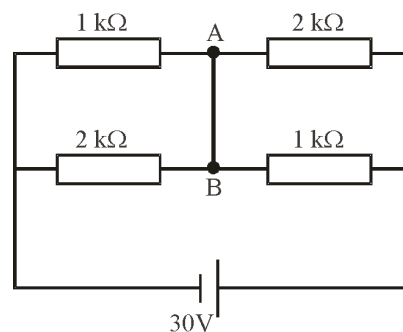


- (1) g, g (2) $\frac{g}{4}, \frac{g}{4}$ (3) $g, \frac{g}{4}$ (4) $\frac{g}{4}, g$

- An ice cube is kept on an inclined plane of angle 30° . The coefficient of kinetic friction between the ice cube and incline plane is $\frac{1}{\sqrt{3}}$. What is the acceleration of the block ?
 (1) Zero (2) 2 m/s^2
 (3) 1.5 m/s^2 (4) 5 m/s^2
- If momentum of body is doubled, then find out how many times kinetic energy will increase ?
 (1) 4 times (2) 3 times
 (3) 2 times (4) 8 times
- Find the net force acting on 2 kg block in given figure?



- (1) 20 kg (2) 10 N (3) 60 N (4) 80 N
- A potential difference of 30V is applied across a color coded carbon resistor with first, second and third rings of blue, black and yellow colors. What is the current flowing through the resistor:-
 (1) $5 \times 10^{-4} \text{ A}$
 (2) $5 \times 10^4 \text{ A}$
 (3) $0.5 \times 10^{-4} \text{ A}$
 (4) $0.5 \times 10^4 \text{ A}$
 - Find the current (in mA) in the wire between points A and B.



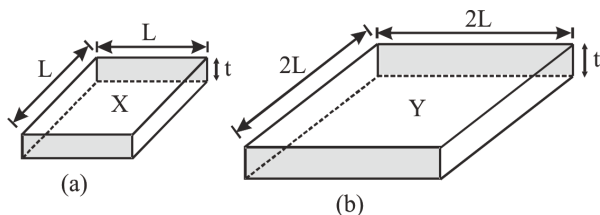
- (1) 2.5 mA (2) 5.0 mA
 (3) 7.5 mA (4) 10.0 mA

English

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12. Figure shows two squares, X and Y, cut from a sheet of metal of uniform thickness t . X and Y have sides of length L and $2L$, respectively :-



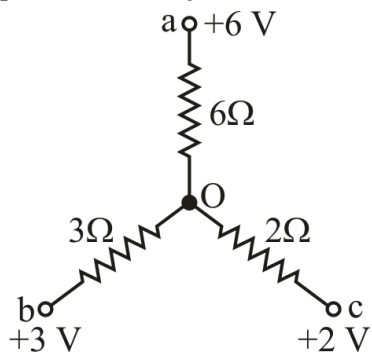
The resistances R_x and R_y of the squares are measured between the opposite faces shaded in Figure. What is the value of R_x/R_y ?

- (1) $1/4$ (2) $1/2$ (3) 1 (4) 2
13. Equivalent resistance between A and B is :-



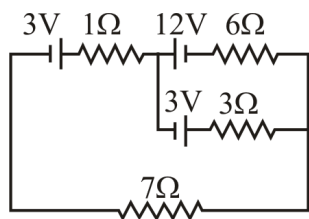
- (1) $5R$ (2) $2R$ (3) $\frac{6R}{5}$ (4) R

14. Find the potential of the junction O.



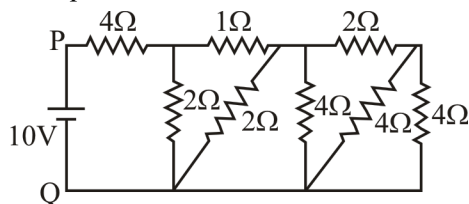
- (1) 3V (2) 2V (3) 5V (4) 10V

15. For the circuit shown, find the current through the 7Ω resistor.



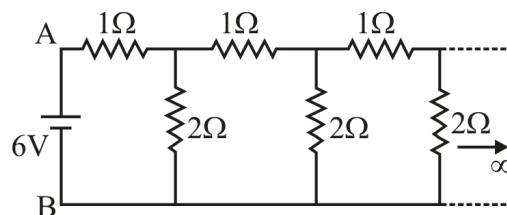
- (1) 0.01A (2) 0.1A (3) 0.001A (4) 1A

16. Find the equivalent resistance between P and Q.



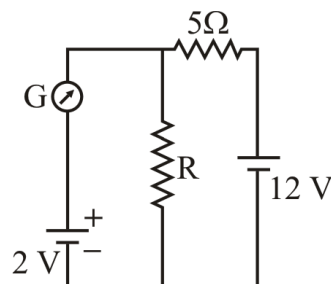
- (1) 10Ω (2) 5Ω (3) 15Ω (4) 20Ω

17. An infinite ladder network is constructed with 1Ω and 2Ω resistors as shown. The effective resistance between A and B is :



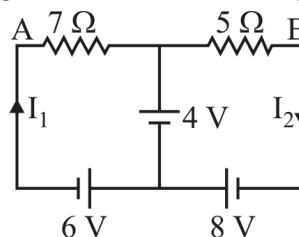
- (1) 1Ω (2) 2Ω (3) 3Ω (4) 4Ω

18. For what value of R is the current in galvanometer zero.



- (1) 5Ω (2) 2Ω (3) 7Ω (4) 1Ω

19. In the circuit given, which of the following is incorrect ?



- (1) Current I_1 is $-\frac{10}{7}A$ (2) Current I_1 is $\frac{10}{7}A$
 (3) Current I_2 is $\frac{12}{5}A$ (4) $V_{AB} = 2V$

20. The resistance of a wire at $20^\circ C$ is 20Ω and at $500^\circ C$ is 60Ω . At which temperature resistance will be 25Ω

- (1) $50^\circ C$ (2) $60^\circ C$ (3) $70^\circ C$ (4) $80^\circ C$

21. If a copper wire is stretched to make it 0.1% longer, the percentage increase in resistance will be :-

- (1) 0.2 (2) 2 (3) 1 (4) 0.1

22. A thin prism of angle 15° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of glass of refractive index 1.75. The combination of the prisms produces dispersion without deviation. The angle of second prism should be :-

- (1) 5° (2) 7° (3) 10° (4) 12°

23. The frequency of a light wave in a material is 4×10^{14} Hz and wavelength is $5000 \text{ \AA}$. The refractive index of material will be :

(1) 1.33 (2) 1.40 (3) 1.50 (4) 3.00

24. A biconvex lens with equal radii of curvature has refractive index 1.6 and focal length 10 cm. Its radius of curvature will be:

(1) 20 cm (2) 16 cm

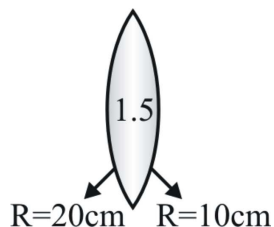
(3) 10 cm (4) 12 cm

25. If the central portion of a convex lens is wrapped in black paper as shown in the figure :-



- (1) No image will be formed by the remaining portion of the lens
 (2) The full image will be formed but it will be less bright
 (3) The central portion of the image will be missing
 (4) There will be two images each produced by one of the exposed portions of the lens

26. Find the power of given lens :-

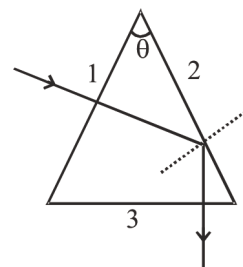


- (1) +5 D (2) +7.5 D
 (3) +2.5 D (4) -2.5 D

27. The magnifying power of the objective of a compound microscope is 7. If the magnifying power of the microscope is 35 then the magnifying power of eye-piece will be :-

(1) 245 (2) 5 (3) 28 (4) 42

28. A ray of light enters into the prism normally with the face 1 and comes out from the face 3. The minimum refractive index of the prism is :-



- (1) $\operatorname{cosec} \theta/2$ (2) $\tan \theta$
 (3) $\sec \theta$ (4) $\operatorname{cosec} \theta$

29. A medium has $n_v = 1.56$, $n_r = 1.44$ then its dispersive power is :

- (1) $\frac{3}{50}$ (2) $\frac{6}{25}$
 (3) 0.03 (4) None of these

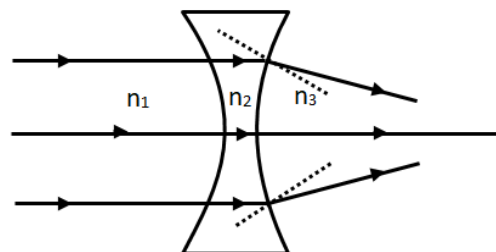
30. A person can't see the objects clearly placed at a distance more than 40 cm. He is advised to use a lens of power -

- (1) +2.5 D (2) -2.5 D
 (3) +0.4 D (4) -0.4 D

31. A convex lens of power 4D & a concave lens of power 3D are placed in contact, the equivalent power of the combination is :

- (1) 1D (2) $\frac{3}{4}$ D
 (3) 7D (4) $\frac{4}{3}$ D

32. From the given figure, establish a relation between n_1 , n_2 , & n_3 :



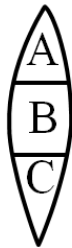
- (1) $n_2 > n_3 > n_1$
 (2) $n_2 < n_3 < n_1$
 (3) $n_3 > n_2$; $n_2 = n_1$
 (4) $n_3 > n_1$; $n_3 = n_2$

English

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33. A thin symmetrical double convex lens of power P is cut into three parts, as shown in the figure. Power of A is :-



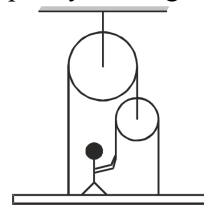
- (1) $2P$ (2) $\frac{P}{2}$ (3) $\frac{P}{3}$ (4) P
34. A lens behaves as a converging lens in air & a diverging lens in water. The refractive index of the lens material is : (refractive index of water = 1.33)
- (1) Equal to unity (2) Equal to 1.33
(3) Between unity & 1.33 (4) Greater than 1.33
35. Convex lens is used in :
- (1) Simple microscope
(2) Telescope
(3) Compound microscope
(4) All of the above

SECTION-B (PHYSICS)

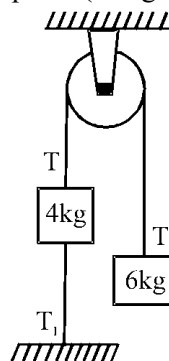
TG: @Chalnaayaaar

36. A planet revolves about the sun in elliptical orbit. The areal velocity $\left(\frac{dA}{dt}\right)$ of the planet is $2 \times 10^{16} \text{ m}^2/\text{sec}$. If the least distance between planet and the sun is $2 \times 10^{12} \text{ m}$ then maximum speed of the planet in km/sec is given by :-
- (1) 20 (2) 60
(3) 40 (4) 100
37. A point mass m is placed inside a spherical shell of radius R and mass M at a distance $\frac{R}{2}$ from the centre of the shell. The gravitational force exerted by the shell on the point mass is :-
- (1) $\frac{GMm}{R^2}$ (2) $\frac{2GMm}{R^2}$
(3) zero (4) $\frac{GMMm}{R^2}$

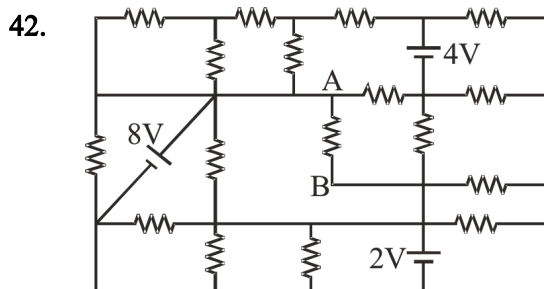
38. In the given diagram, with what force must the man pull the rope to hold the plank in position? Mass of the man is 80 kg. Neglect the weights of plank, rope and pulley. Take $g = 10 \text{ ms}^{-2}$



- (1) 200 N (2) 300 N (3) 600 N (4) 150 N
39. Two bodies of masses 4 kg and 6 kg are attached to the ends of a string passing over a pulley. The 4 kg mass is attached to the table top by another string. The tension in this string T_1 is equal to (take $g = 10 \text{ ms}^{-2}$) :-



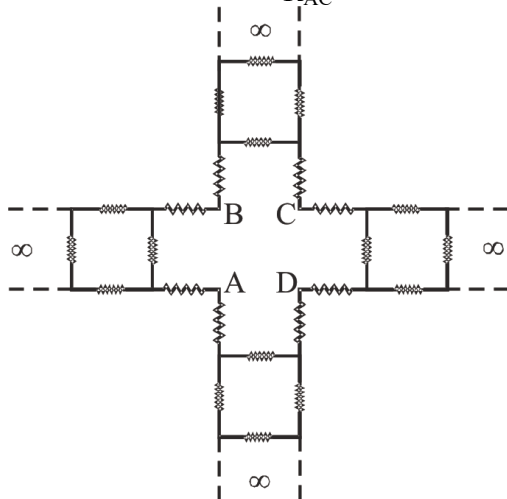
- (1) 20 N (2) 25 N (3) 10.6 N (4) 10 N
40. The ratio of weights of a man in a stationary lift and in a lift accelerating downward with a uniform acceleration 5 : 4. The acceleration of the lift is ?
- (1) $\frac{g}{3}$ (2) $\frac{g}{2}$ (3) g (4) $\frac{g}{5}$
41. When 10 wagons of trains are pulled by force of 10^5 , if mass of each wagon is 1500 kg then find force exerted on last four wagon ?
- (1) $2 \times 10^4 \text{ N}$ (2) $4 \times 10^4 \text{ N}$
(3) $3 \times 10^4 \text{ N}$ (4) $5 \times 10^4 \text{ N}$

Each resistance is 6Ω

Find current in AB

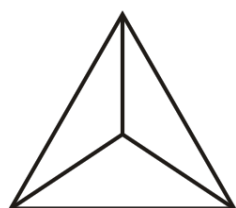
- (1) 1.3A (2) 1A (3) 2A (4) 1.5A

43. Four infinite ladder network contain identical resistance of $R\Omega$ as shown. The equivalent resistance between A & B is R_{AB} and between A & C is R_{AC} . then the value of $\frac{R_{AB}}{R_{AC}}$ is :-



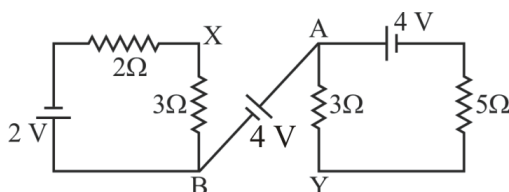
- (1) $3/4$ (2) $4/3$ (3) 2 (4) $1/2$

44. Six equal resistances, each 4Ω , make the arms of the given figure. The effective resistance between any two corners is :-



- (1) 10Ω (2) $\frac{2}{3}\Omega$
(3) 4Ω (4) 2Ω

45. In the given circuit find the potential difference between X and Y.



- (1) 6.7 V (2) 3.7 V
(3) 4 V (4) 10 V

46. Masses of three wires of same material are in the ratio 1 : 3 : 5 and their lengths are in the ratio 5 : 3 : 1. The ratio of their electrical resistance is :-

- (1) 1 : 3 : 5 (2) 5 : 3 : 1
(3) 1 : 15 : 125 (4) 125 : 15 : 1

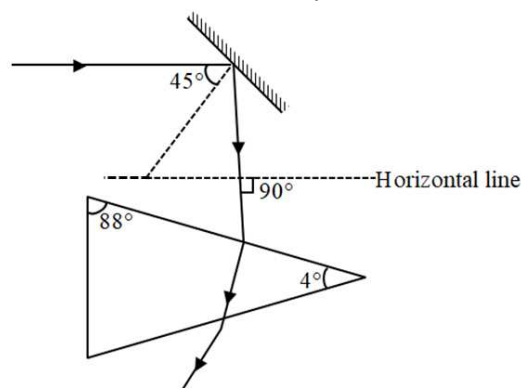
47. An astronomical telescope has an angular magnification of magnitude 5 for distant objects. The separation between the objective and eye-piece is 36 cm and the final image is formed at infinity. Determine the focal length of objective and eye-piece-

- (1) $f_0 = 30$ cm, $f_e = 6$ cm
(2) $f_0 = 25$ cm, $f_e = 10$ cm
(3) $f_0 = 30$ cm, $f_e = 10$ cm
(4) $f_0 = 15$ cm, $f_e = 5$ cm

48. A small angle prism ($\mu = 1.62$) gives a minimum deviation of 4.8° . The angle of prism is :-

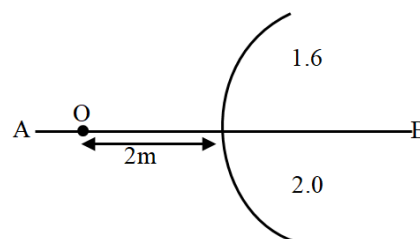
- (1) 7.74° (2) 5° (3) 6.36° (4) 3°

49. A ray of light strikes a plane mirror at an angle of incidence 45° as shown. After reflection, the ray passes through a prism of $n = 1.5$ & $A = 4^\circ$. The angle through which the mirror should be rotated, if the total deviation of the ray is to be 90° , is :



- (1) 1° clockwise (2) 1° anticlockwise
(3) 2° clockwise (4) 2° anticlockwise

50. In the figure shown, a point object O is placed in air. A spherical boundary of radius of curvature 1.0 m separates two media. AB is principal axis. The refractive index above AB is 1.6 & below AB is 2.0. The separation between the images formed due to refraction at spherical surface is :

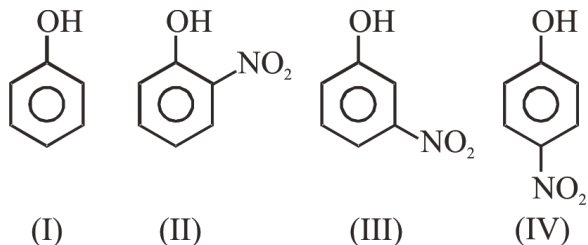


- (1) 12 m (2) 20 m (3) 14 m (4) 10 m

Topic : GOC-I (ACIDIC STRENGTH TO TAUTOMERISM), GOC-II, ELECTROCHEMISTRY [BACK UNIT]
: CHEMICAL KINETICS TG: @Chalnaayaaar

SECTION-A (CHEMISTRY)

51. Correct order of K_a value for given compounds



(1) $IV > II > III > I$

(2) $III > IV > II > I$

(3) $IV > III > II > I$

(4) $IV > III > I > II$

52. CHCl_3 is more acidic than CHF_3 because :-

- (1) Cl has more -I than that of F
- (2) Cl has vacant d-orbital, therefore conjugate base of CHCl_3 is resonance stabilized
- (3) CHCl_3 has more $\text{p}K_a$ than CHF_3
- (4) Both (2) & (3)

53. which H is abstracted easily :-

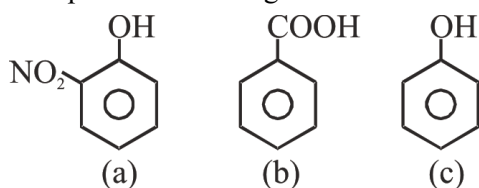
(1) α

(2) β

(3) γ

(4) with equal ease

54. Compare acidic strength :-



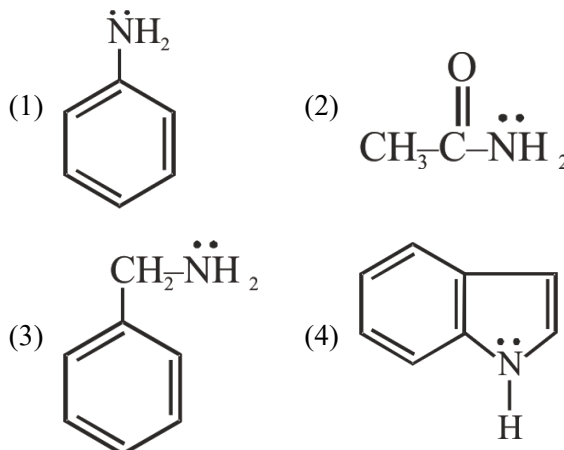
(1) $a > b > c$

(2) $b > c > a$

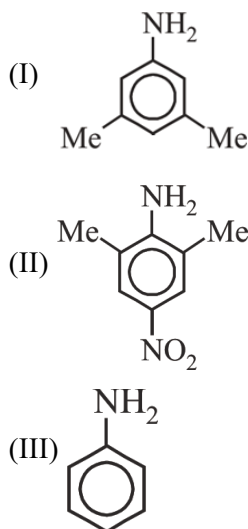
(3) $b > a > c$

(4) $a > c > b$

55. Which of the following compound is most basic:-



56. Which of the following orders is correct for the basic strength of these anilines ?



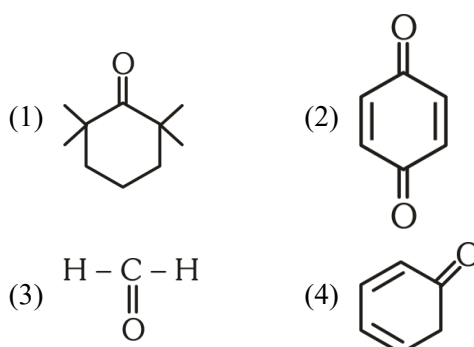
(1) $I > III > II$

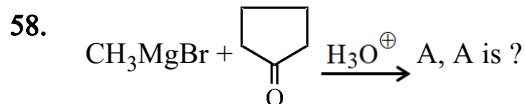
(2) $III > II > I$

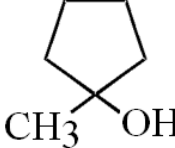
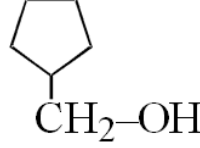
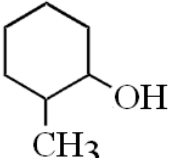
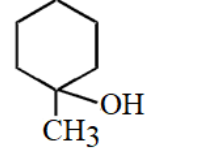
(3) $III > I > II$

(4) $II > I > III$

57. Which of the following compound can show tautomerism ?



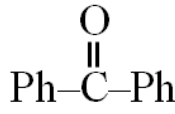
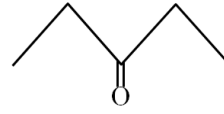
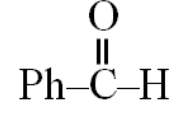
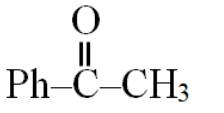


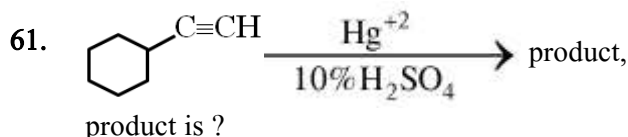
- (1)  (2) 
 (3)  (4) 

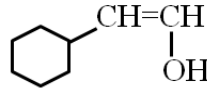
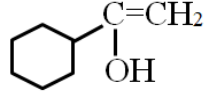
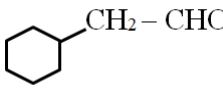
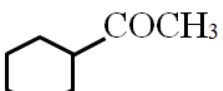
59. Which of the following will form racemic mixture on adding HCN ?

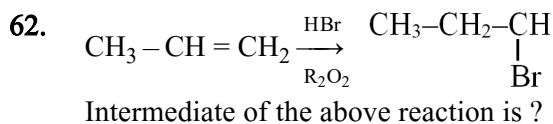
- (1) Formaldehyde (2) Acetaldehyde
 (3) Benzophenone (4) Acetone

60. Which of the following compound will react with NaHSO_3 ?

- (1)  (2) 
 (3)  (4) 



- (1) 
 (2) 
 (3) 
 (4) 



- (1) $\text{CH}_3 - \overset{\oplus}{\text{CH}} - \text{CH}_3$ (2) $\text{CH}_3 - \underset{\text{Br}}{\underset{\cdot}{\text{CH}}} - \text{CH}_2$
 (3) $\text{CH}_3 - \text{CH}_2 - \overset{\oplus}{\text{CH}_2}$ (4) $\text{CH}_3 - \underset{\text{Br}}{\underset{\cdot}{\text{CH}}} - \text{CH}_2$

63. If $r = k [\text{A}]_0^2 [\text{B}]_0$ for a reaction, by what factor is the initial rate multiplied if the $[\text{A}]_0$ is multiplied by 1.5 and the $[\text{B}]_0$ is tripled ?

- (1) 4.5 (2) 2.25
 (3) 6.75 (4) None of these

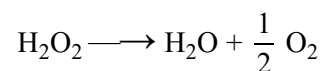
64. For the reaction $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$, the rate of disappearance of H_2 is $1.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. The rate of appearance of HI will be :-

- (1) $1.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (2) $0.50 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (3) $2.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
 (4) $4.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

65. For an elementary reaction, $\text{X}(\text{g}) \rightarrow \text{Y}(\text{g}) + \text{Z}(\text{g})$ the half life period is 10 min. In what period of time would be the concentration of X be reduced to 10% of original concentration ?

- (1) 20 Min. (2) 33 Min.
 (3) 15 Min. (4) 25 Min.

66. Which rate equation is correct for reaction



- (1) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = \frac{1}{2} \frac{d[\text{O}_2]}{dt}$
 (2) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{1}{2} \frac{d[\text{H}_2\text{O}]}{dt} = \frac{d[\text{O}_2]}{dt}$
 (3) $-\frac{1}{2} \frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = 2 \frac{d[\text{O}_2]}{dt}$
 (4) $-\frac{d[\text{H}_2\text{O}_2]}{dt} = \frac{d[\text{H}_2\text{O}]}{dt} = 2 \frac{d[\text{O}_2]}{dt}$

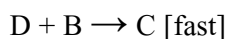
67. Differential form of the rate equation is :

$$\frac{dx}{dt} = k[P][Q]^{0.5}[R]^{0.5}$$

which statement about the above equation is wrong ?

- (1) Order with respect to P is one
- (2) Total order of the reaction is two
- (3) Order with respect to each of Q and R is 0.5
- (4) Unit of specific reaction rate is mol s^{-1}

68. The mechanism of the reaction : $A + B \rightarrow C$ is as under :



Choose the set showing rate equation and order with respect to A and B

- (1) $k_1[A]$, 1,0
- (2) $k_1[A]^2$, 0,1
- (3) $k_1[A][B]$, 2,0
- (4) $k_1[A]^2$, 2,0

69. For a given reaction, the overall enthalpy change is +100 kJ/mol and activation energy of the reverse reaction is +200 kJ/mol. The activation energy for the forward reaction would be

- (1) +300 kJ/mol
- (2) +200 kJ/mol
- (3) +0.100 kJ/mol
- (4) +100 kJ/mol

70. The $t_{0.5}$ for the first order reaction, $\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ is 20 min. The time in which the conc. of PCl_5 reduces to 25% of the initial conc. is close to-

- (1) 22 min
- (2) 40 min
- (3) 90 min
- (4) 50 min

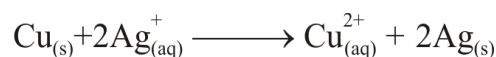
71. In an electrochemical cell that function as a voltaic cell :-

- (1) electrons move from the cathode to the anode
- (2) electrons move through a salt-bridge
- (3) electrons can move either from the cathode to the anode or from the anode to the cathode
- (4) reduction occurs at the cathode

72. The weight ratio of Al and Ag deposited using the same quantity of current is :-

- (1) 9 : 108
- (2) 2 : 12
- (3) 108 : 9
- (4) 3 : 8

73. Which of the following conditions will increase the voltage of the cell represented by the equation:



- (1) Increase in the concentration of Ag^{+} ions
- (2) Increase in the concentration of Cu^{2+} ions
- (3) Increase in the size of silver electrode
- (4) Increase in the size of copper electrode

74. (i) $\text{Cu} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2(\text{g})$;

$$E^{\circ}_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$$

(ii) $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$;

$$E^{\circ}_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$$

(iii) $\text{Ag} + 2\text{HCl} \rightarrow \text{AgCl} + \text{H}_2(\text{g})$;

$$E^{\circ}_{\text{Ag}^{+}/\text{Ag}} = +0.80 \text{ V}$$

Which of the above reactions is feasible ?

- (1) (ii)
- (2) (i)
- (3) (iii)
- (4) All

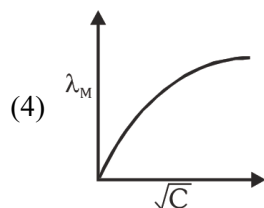
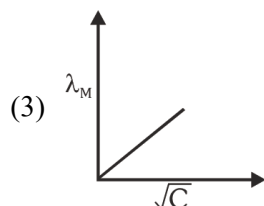
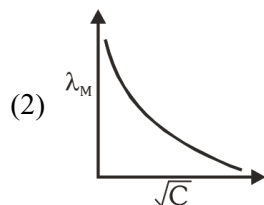
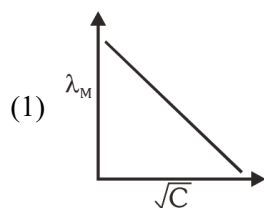
75. What will be the molar conductance at infinite dilution for NH_4OH , if at infinite dilution the molar conductances of $\text{Ba}(\text{OH})_2$, BaCl_2 and NH_4Cl are 523.28, 280.0 and $129.8 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively.

- (1) 502.88
- (2) 373.08
- (3) 251.44
- (4) 226.96

76. The ionic conductivity of Ba^{+2} and Cl^- at infinite dilution are 127 and $76 \text{ ohm}^{-1} \text{ cm}^2 \text{ eq}^{-1}$ respectively. The molar conductivity of BaCl_2 at infinite dilution is :-

- (1) $203 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (2) $406 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (3) $279 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (4) $127 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$

77. Which of the following curve represents the variation of λ_M with $\sqrt{C}$ for AgNO_3 ?



78. For the cell
 $\text{Zn}_{(s)} | \text{Zn}^{+2}(10^{-2}\text{M}) || \text{Zn}^{+2}(10^{-1}\text{M}) | \text{Zn}_{(s)}$

- (1) $E_{\text{cell}} > 0$ and $\Delta G < 0$
- (2) $E_{\text{cell}}^{\circ} = 0$ and $\Delta G > 0$
- (3) $E_{\text{cell}}^{\circ} = 0$ and $\Delta G^{\circ} = 0$
- (4) Both (1) and (3)

79. Products obtained at cathode and anode during the electrolysis of dilute $\text{NaCl}_{(\text{aq})}$ solution.

- (1) H_2 and Cl_2
- (2) H_2 and O_2
- (3) Cl_2 and H_2
- (4) O_2 and H_2

80. Which of the following relation is correct ?

- (1) $\kappa = \frac{G}{G^*}$
- (2) $\lambda_m = \frac{\kappa \times 1000}{N}$
- (3) $G^* = \frac{\ell}{A}$
- (4) $\lambda_{\text{eq}} = \lambda_m \times \text{V.F.}$

81. Which of the following is correct for the cell
 $\text{Cr} | \text{Cr}^{+3} (0.01\text{M}) || \text{Cr}^{+3} (0.01\text{M}) | \text{Cr}$

- (1) $E_{\text{cell}}^{\circ} = 0$
- (2) $E_{\text{cell}} = 0$
- (3) Cr is oxidised to Cr^{+3} at anode
- (4) All of these

82. Select the correct statement among the following :

- (1) order of reaction may not be fraction
- (2) molecularity of a reaction may be zero
- (3) zero order reaction may be complex as well elementary
- (4) 2nd order reaction may be elementary as well complex

83. For a first order reaction which of the following relation is correct ?

- (1) $kt = 2.303 \log \left(\frac{A_0}{A_t} \right)$
- (2) $t_{1/2} = \frac{0.693}{k}$
- (3) $t_{1/2} \propto A_0$
- (4) All are correct

84. Which of the following does not depend on catalyst

- (1) E_a (Activation energy)
- (2) ΔH
- (3) Rate of reaction
- (4) Rate constant (k)

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85. For a first order reaction unit of Arrhenius factor (A) will be

- (1) sec^{-1} (2) min^{-1}
 (3) both (1) & (2) (4) $\text{mol l}^{-1} \text{sec}^{-1}$

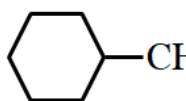
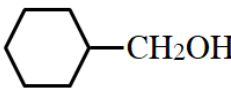
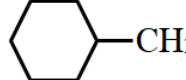
SECTION-B (CHEMISTRY)

86. Which of the following has highest enol content.

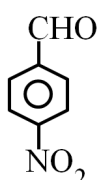
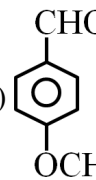
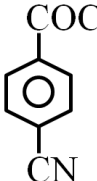
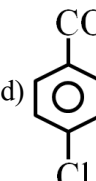
- (1) $\text{CH}_3-\text{C}(=\text{O})-\text{H}$
 (2) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$
 (3) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{C}(=\text{O})-\text{OC}_2\text{H}_5$
 (4) $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3$

87. $\text{B} \xleftarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{BH}_3, \text{THF}} \text{Cyclohexene} \xrightarrow[\text{NaBH}_4]{\text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}} \text{A}$

A and B are respectively :-

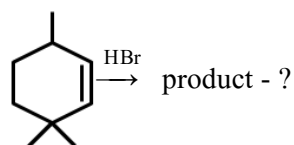
- (1)  both
 (2)  both
 (3)  and 
 (4)  and 

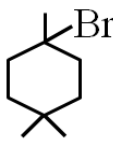
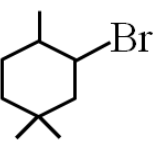
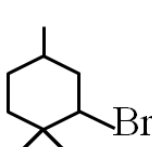
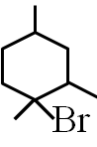
88. Reactivity towards nucleophilic addition reaction ?

- (a)  (b) 
 (c)  (d) 

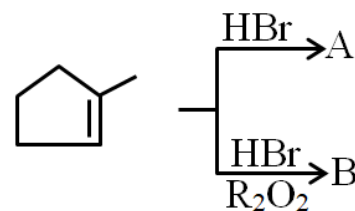
- (1) $a > c > d > b$ (2) $a > c > b > d$
 (3) $a > b > c > d$ (4) $a > b > d > c$

89.

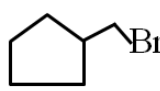
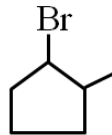
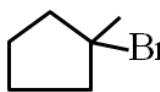
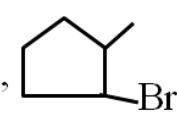
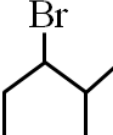
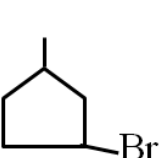
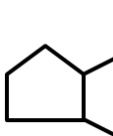
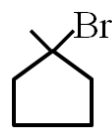


- (1)  (2) 
 (3)  (4) 

90.

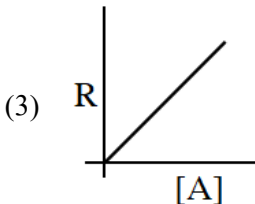


A and B respectively are

- (1)  , 
 (2)  , 
 (3)  , 
 (4)  , 

91. Select correct statement

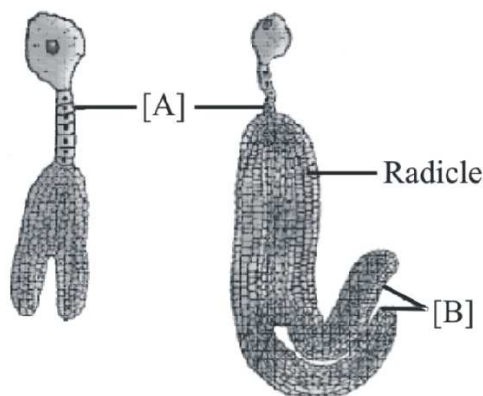
- (1) Benzaldehyde is more reactive towards NAR than Propanal
 (2) The position of equilibrium lies largely to the left hand side on addition of NaHSO_3
 (3) Aldehydes react with one equivalent of monohydric alcohol to form acetals.
 (4) Butanone forms racemic mixture on adding HCN.

92. If a 1st order reaction is one fifth completed in 40 min, the time required for its 100% completion is :-
- 100 min
 - 200 min
 - 350 min
 - Infinite
93. 80% of a first order reaction completes in 70 min. How much time it will take to complete 90% of reaction ?
- 114 min.
 - 140 min.
 - 100 min.
 - 200 min.
94. Same quantity of current is passed through molten NaCl and molten Al₂O₃. If 4.6 g of Na was deposited in one cell, the mass of Al deposited in other cell is :-
- 0.9 g
 - 1.8 g
 - 2.7 g
 - 3.6 g
95. How many electrons are delivered at the cathode during electrolysis by a current of 1 amp in 60 seconds?
- 3.74×10^{20}
 - 6.0×10^{23}
 - 7.48×10^{21}
 - 6.0×10^{20}
96. Reactivity of metals increases as electropositivity increases, what will happen if a copper spoon is used to stir a solution of aluminium nitrate?
- The spoon will get coated with aluminium
 - An alloy of aluminium and copper is formed
 - The solution becomes blue
 - There is no reaction.
97. The emf of the cell,
 $\text{Ni}|\text{Ni}^{2+}(1.0 \text{ M})||\text{Ag}^+(1.0 \text{ M})|\text{Ag}$ will be :
 (E° of $\text{Ni}^{2+}|\text{Ni} = -0.25$ volt, E° for $\text{Ag}^+|\text{Ag} = 0.80$ volt)
- 0.55 volt
 - 1.05 volt
 - +1.05 volt
 - 0.55 volt
98. Ratio of moles of Na, Mg and Al deposited at electrodes if equal amount of charge is passed through cells.
- 1 : 1 : 1
 - 23 : 24 : 27
 - 1 : 2 : 3
 - 6 : 3 : 2
99. During electrolysis of molten NaCl if 46 g of Na is deposited at cathode, amount of Cl₂ liberated at anode will be
- 23 g
 - 35.5 g
 - 71 g
 - 46 g
100. For the reaction following rate law as $R = k[A]^2$ which of the following relation is correct :-
- Rate of reaction becomes double on increasing the concentration up to 2 times
 - $t_{1/2} \propto \frac{1}{A_0}$
 - 
 - $t_{1/2}$ is independent of initial concentration of A.

Topic : PRINCIPLES OF INHERITANCE (POI) COMPLETE, LOA : POST FERTILIZATION TO LAST ECOLOGY [BACK UNIT] : HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-I)

101. In the following figures, (A) and (B) represent respectively



- (1) Plumule and cotyledons
 (2) Cotyledons and plumule
 (3) Suspensor and cotyledons
 (4) Cotyledons and suspensor
102. Which of the following is an ultimate sign of sexual reproduction ?
- (1) Formation of embryo
 (2) Formation of zygote
 (3) Formation of embryo and endosperm
 (4) Formation of seed
103. The real function of the 'endosperm' is to :-
- (1) Supply nutrition to the growing embryo
 (2) Form integuments of ovule
 (3) Form funicle of ovule
 (4) Supply nutrient to embryosac
104. In a type of apomixis known as adventive embryony, embryos develop directly from the :-
- (1) Nucellus or integuments
 (2) Synergids or antipodals in an embryo sac
 (3) Accessory embryo sacs in the ovule
 (4) Zygote

105. Polyembryony seen in :

- (1) *Citrus*
 (2) Coconut
 (3) Date palm
 (4) *Yucca*

106. Species which play, a vital role in controlling the relative abundance of other species, and control functioning of the community is called ?

- (1) Dominant species
 (2) Key stone species
 (3) Link species
 (4) Network species

107. Match the column-I with column-II

	Column-I		Column-II
(i)	Endemic species	(a)	Kangaroo rat
(ii)	Keystone species	(b)	<i>Mycorrhizal</i> fungi
(iii)	Critical link species	(c)	Kangaroo

- (1) i-a, ii-b, iii-c
 (2) i-c, ii-a, iii-b
 (3) i-c, ii-b, iii-a
 (4) i-b, ii-a, iii-c
108. Select true statement :-
- (1) Different population of different species are called sister population found in different area.
 (2) Population is basic unit of Ecological study
 (3) Community is the different populations of same species
 (4) Local population is also known as demes

109.

	Column-I		Column-II		Column-III
(1)	Species diversity	(a)	Great influence on community characters	(i)	Maximum species diversity in tropical rain forest
(2)	Species dominance	(b)	Vertical layering according to need of light	(ii)	<i>Prosopis</i> in Aravalli hills
(3)	Stratification	(c)	Different type of species in a community	(iii)	Littoral zone, Limnetic zone & Profundal zone
(4)	Keystone species	(d)	Highest no. of organism of a species	(iv)	<i>Pisaster</i> star fish

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- (1) 1-c-i, 2-d-ii, 3-b-iii, 4-a-iv
 (2) 1-a-iv, 2-b-iii, 3-c-ii, 4-d-i
 (3) 1-c-iii, 2-b-iv, 3-c-ii, 4-d-i
 (4) 1-b-ii, 2-d-iii, 3-c-i, 4-d-iv

110. Which of the following statement is incorrect :-

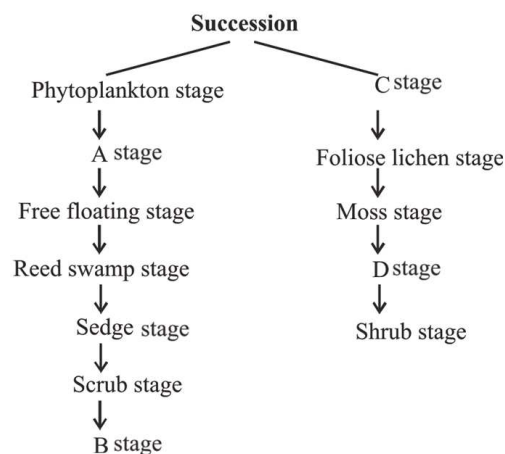
- (1) Study of the relation of the group of different species with their environment is known as synecology.
 (2) Different population of same species of organisms which are found in different places are known as metapopulation.
 (3) A species which is found only in a particular area is known as endemic species.
 (4) Biotic communities are never stable.

111.

Column-I	Column-II	Column-III
(1) Primary succession	(a) Occurs in burnt forest	(i) Pioneer community- <i>Crustose Lichen</i>
(2) Secondary succession	(b) Occurs on barren rock	(ii) Slow process
(3) Hydrosere	(c) Succession on rock	(iii) Pioneer community-Phyto-planktons
(4) Lithosere	(d) Newly formed pond	(iv) Fast process

- (1) 1-b-ii, 2-a-iv, 3-d-iii, 4-c-i
 (2) 1-a-iv, 2-b-iii, 3-c-ii, 4-d-i
 (3) 1-d-i, 2-c-ii, 3-c-iii, 4-d-iv
 (4) 3-a-iii, 2-b-ii, 1-d-i, 4-c-iv

112.



Which of the following is correct ?

- (1) A-Rooted submerged, B-forest, C-Crustose lichen, D-*Typha*
 (2) A-Herb, B-forest, C-Rooted submerged, D-Herb
 (3) A-Rooted submerged, B-forest, C-Crustose lichen, D-Herb
 (4) A-Rooted submerged, B-moss, C-Crustose lichen, D-Shrub

113. Which of the following statements is/are not correct ?

- (A) In succession, community changes in orderly and sequential manner.
 (B) As succession proceeds, numbers and types of decomposers also changes.
 (C) The community that invade a barren area, called as climax community.
 (D) Both xerarch and mesarch lead to a hydrarch condition.
- (1) A and B
 (2) A, B and C
 (3) B, C and D
 (4) C and D

114. Secondary succession is development of biotic communities on :-
 (a) Newly exposed habitat
 (b) Cleared forest area
 (c) Freshly harvested crop field
 (d) Pond filled after a dry season
 (1) only a
 (2) a & d
 (3) a, c, d
 (4) b,c,d
115. Select the example of mutualism :-
 (1) Shark and sucker
 (2) Crocodile and birds
 (3) Copepods and marine fish
 (4) *Yucca* plant and moth
116. Co-evolution, Co-existence and Co-extinction, all occur in :-
 (1) Commensalism
 (2) Mutualism
 (3) Competition
 (4) Amensalism
117. **Assertion** : Hybrid seeds have to be produced every year because seeds collected from hybrid plants, if sown subsequently, do not maintain hybrid characters.
Reason : Hybrid seeds show segregation of traits.
 (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) Assertion is True but the Reason is False.
 (4) Both Assertion & Reason are False.
118. Which of the following group do not comes under commensalism?
 (1) *Orchid* growing on Mango branch
 (2) Lichens and fungi
 (3) Cattle egret and grazing cattle
 (4) *Sea Anemone* and clown fish
119. Orchid grows as a _____ on a mango branch.
 (1) Prey
 (2) Parasite
 (3) Epiphyte
 (4) Predator
120. Choose the mismatch set :-
 (1) Lichens - Mutualism
 (2) *Barnacles* on whale - Commensalism
 (3) *Mycorrhiza* - Mutualism
 (4) *Plasmodium* & Human - Predation
121. If the female bee's colour pattern changes, the orchid flower co-evolves to maintain the resemblance of its petals for the successful pollination is known as :-
 (1) Commensalism
 (2) Protocooperation
 (3) Co-evolution
 (4) Predation
122. Which one of the following occur as a pioneer species on rocks ?
 (1) Bryophytes
 (2) Phytoplanktons
 (3) Lichens
 (4) Herbs

123. Match the column I with column II :

	Column-I (Interactions)		Column-II (Sign)
(A)	Predation	(i)	+, +
(B)	Commensalism	(ii)	-, 0
(C)	Mutualism	(iii)	+, -
(D)	Amensalism	(iv)	+, 0

- (1) A-iii, B-iv, C-i, D-ii
 (2) A-iii, B-i, C-iv, D-ii
 (3) A-i, B-ii, C-iii, D-iv
 (4) A-iv, B-iii, C-ii, D-i
124. Which one of the following is not the part of hydrarch succession ?
 (1) Scrub stage
 (2) Tree stage
 (3) Moss stage
 (4) Submerged plant stage
125. In which interaction one species benefits and other is neither harmed nor benefited ?
 (1) Mutualism
 (2) Predation
 (3) Commensalism
 (4) Amensalism
126. **Statement-I** : Female wasp uses the fruit of fig only as an oviposition site.
Statement-II : A given fig species can be pollinated by both bees and bumblebees species.
 (1) Both statement I and statement II are correct
 (2) Statement I is correct and statement II is incorrect
 (3) Statement I is incorrect and statement II is correct
 (4) Both statement I and statement II are incorrect

127. 'Clonal selection' refers to ?

- (1) Specific cloning of T-helper cells.
 (2) Specific cloning of T-killer and B-cells.
 (3) Non-specific cloning of T-helper cells.
 (4) Non-specific cloning of T-killer and B-cells.

128. All the diseases are spread by mosquito except :-

- (1) Chikungunya (2) Dengue
 (3) Chicken pox (4) Filariasis

129. Which of the following drug is obtained from *Claviceps purpurea* ?

- (1) Charas (2) *Belladonna*
 (3) LSD (4) Heroin

130. Principle of vaccination is based on :-

- (1) Discrimination between self-non self
 (2) Diversity
 (3) Memory
 (4) Specificity

131. Bluish coloration of lips & nails due to infection of alveoli may be caused due to :-

- (1) orthomyxo virus
 (2) *Haemophilus influenza*
 (3) rhino virus
 (4) *Carnibacterium diptheriae*

132. In life cycle of Plasmodium gametocytes are formed in :-

- (1) Gut of mosquito
 (2) Salivary gland of mosquito
 (3) Liver of human
 (4) RBCs of human

133. How many of the following symptoms are associated with Ascariasis-

Muscular pain, Excessive mucous in stool, fever, Anemia, Intestinal blockage, dysentery :

- (1) 2 (2) 3 (3) 4 (4) 5

134. How many of the following is/are mode of transmission of AIDS ?

- (a) Sexual contact with infected person.
- (b) By sharing food
- (c) Transfusion of contaminated blood
- (d) By sharing infected needles
- (e) Mosquito bites

- (1) Five (2) Four
- (3) Two (4) Three

135. Cancer cells do not exhibit the property of :-

- (1) Generating tumors
- (2) Metastasis
- (3) Contact inhibition
- (4) Less number of mitochondria

SECTION-B (BIOLOGY-I)

136. What would be the number of chromosomes in the cells of the aleurone layer in a plant species with 8 chromosomes in its synergids ?

- (1) 16 (2) 24 (3) 32 (4) 8

137. Select the incorrect statement from the following?

- (1) *Asteraceae* shows apomixis
- (2) Mango shows polyembryony
- (3) Portion below the level of cotyledon is hypocotyl
- (4) Grass family have 2 scutellum in their seed

138. Ecology at the organismic level is essentially physiological ecology which tries to understand that how :-

- (1) Ecosystems interact among themselves.
- (2) Different population interact and comprise a community.
- (3) Different organisms adapted to their environment in terms of not only survival but also reproduction
- (4) Different ecosystem forms a biome.

139. About succession which of the following statement is correct :-

- (1) In xerosere, xeric conditions progress to hydric conditions
- (2) In hydrosere, mesic environment progress to hydric conditions
- (3) In hydrosere, hydric environment progress to mesic conditions
- (4) Abandoned farm lands show primary succession

140. Choose the incorrect match :-

	Column-I (Community characteristic)	Column-II (Seral stages)	Column-III (Climax stage)
(i)	Species diversity	Less	More
(ii)	Niches	Few	Many
(iii)	Niche types	Narrow	Broad
(iv)	Food chains and food web	Simple	Complex
(v)	Vegetation	Mesophytes	Xerophytes

- (1) (i) and (ii) only
- (2) (iii) only
- (3) (iii) and (v)
- (4) (i), (iv) and (v)

141. The Mediterranean orchid employs _____ to get pollination done by a species of bee.

- (1) Shelter
- (2) Food
- (3) Sexual deceit
- (4) Egg-laying sites

142. The species that invade a bare area is called ?

- (1) Sere
- (2) Pioneer species
- (3) Climax species
- (4) Key stone species

143. In hydrarch succession, the successional series progress from :-

- (1) Mesic to hydric condition
- (2) Hydric to mesic condition
- (3) Hydric to xeric condition
- (4) Xeric to mesic condition

144. Why secondary succession is faster ?

- (1) Because soil is already there
- (2) They have special power
- (3) Growth of plants is faster
- (4) All of these

145. Smoking is associated with :

- (1) Lung cancer
- (2) emphysema
- (3) Coronary heart disease
- (4) All

146. Which of the following is a pair of viral diseases?

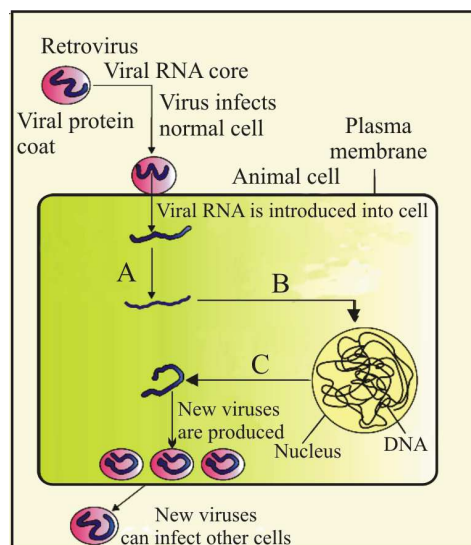
- (1) Typhoid, pneumonia
- (2) Ringworm, AIDS
- (3) Common cold, AIDS
- (4) Dysentery, cancer

147. How many of the following are sexually transmitted diseases ?

AIDS, Trichomoniasis, Syphilis, Chlamydia, Genital herpes, diphtheria, tuberculosis

- | | |
|-----------|----------|
| (1) Three | (2) Six |
| (3) Four | (4) Five |

148. The figure below shows steps A, B & C of replication of retrovirus. Select the option giving correct identification



- (1) B-Incorporation of viral DNA into host genome with integrase enzyme
- (2) C-Production of new viral DNA by infected cell
- (3) A-Replication of viral RNA by protease enzyme
- (4) C-Viral RNA replication by reverse transcriptase enzyme

149. Which of the following statements are not incorrect ?

- (a) Smack is chemically diacetyl morphine which is obtained by acetylation of heroin
- (b) Morphine is obtained from the latex of poppy plant *Papaver somniferum*
- (c) Drugs having their effects on cardiovascular system of body are being abused by some sports persons.
- (d) Crack interferes with the transport of the neurotransmitter dopamine

- | | |
|----------------|-------------|
| (1) a, b, c, d | (2) a, c, d |
| (3) b, c, d | (4) c, d |

150. *Entamoeba histolytica* causes :-

- (1) Dry, scaly lesions on skin
- (2) Cough and headache
- (3) abdominal pain & cramps
- (4) Fever with chills

- 159. Statement-I :-** Mendel selected 14 true-breeding pea plant varieties.
- Statement-II:-** Mendel observed that all F_1 progeny plants had recessive phenotype.
- (1) Both statement I and II are true.
 - (2) Statement I is false, II is true
 - (3) Statement I is true, II is false
 - (4) Both statement I&II are false
- 160.** Select genotypic ratio in mendelian dihybrid cross –
- (1) 9 : 3 : 3 : 1
 - (2) 3 : 1
 - (3) 1 : 2 : 2 : 4 : 1 : 2 : 1 : 2 : 1
 - (4) 1 : 1
- 161.** Shape of seed depends on starch granules size. Inheritance of seed shape show.....relationship while inheritance of starch grains shows.....
- (1) Dominant-recessive, co-dominance
 - (2) Incomplete dominance, co-dominance
 - (3) Dominant-recessive, incomplete dominance
 - (4) Co-dominance, incomplete dominance
- 162.** Which idea is depicted by a cross in which the F_1 generation resembles both the parents ?
- (1) Inheritance of one gene
 - (2) Codominance
 - (3) Incomplete dominance
 - (4) Complete dominance
- 163.** In a diploid organism how many maximum different alleles of a gene are present :-
- (1) One
 - (2) Two
 - (3) Many
 - (4) Variable in different chromosome
- 164.** When more than one phenotype is influenced by the same gene then dominance is not an autonomous feature of a gene. It depends upon :-
- (1) Gene product
 - (2) The particular phenotype that we choose to examine.
 - (3) Number of genes
 - (4) Both 1 and 2
- 165.** Which of the following is example of pleiotropic gene :-
- (1) Seed colour in garden pea
 - (2) Sickle cell anaemia
 - (3) Flower colour in *Mirabilis jalapa*
 - (4) All of the above
- 166.** Which of the following is a mismatch with respect to F_2 phenotypic ratio ?
- (1) Polygenic inheritance - 1:4:6:4:1
 - (2) Dominant epistasis - 12:3:1
 - (3) Codominance - 1:2:1
 - (4) Complementary gene interaction - 9:3:4
- 167.** If the minimum and maximum height of sugarcane plant is 50 cm and 200 cm respectively and this character is regulated by 3 genes. Then find out the height of plants produced in the progeny of cross $aabbCC \times AABBcc$:-
- (1) 75 cm
 - (2) 150 cm
 - (3) 50 cm
 - (4) 125 cm
- 168.** Cytochrome or plasma gene expresses itself in :-
- (1) Only female
 - (2) Only male
 - (3) Male and female both
 - (4) None of the above

169. In *Mirabilis* plant inheritance of leaf colour is due to plastids. If pollen grain of green branch is used to pollinate the stigma of variegated branch flower, the result is :-

- (1) Only green leaves
- (2) Only variegated leaves
- (3) Green, pale or variegated leaves
- (4) Both green and variegated leaves

170. **Assertion (A) :-** Results in Mendel's reciprocal cross were unchanged.

Reason (R) :- Characters studied by Mendel were autosomal.

- (1) Both A & R are true & R is correct explanation of A.
- (2) Both A & R are true but R is not correct explanation of A
- (3) A is correct but R is false
- (4) Both A & R are false.

171. Assume that gene *a* and *b* are linked and show 40% recombination. If ++ / ++ individual is crossed with ab / ab, then types and proportions of gametes in F_1 will be :-

- (1) ++ 20% : ab 20% : +b 20% : a + 40%
- (2) ++ 50% : ab 50%
- (3) ++ 25% : ab 25% : + b 25% : a + 25%
- (4) ++ 30% : ab 30% : + b 20% : a + 20%

172. **Statement-I:-** Sex linkage was discovered by Morgan.

Statement-II:- Linkage was proved by Morgan.

- (1) Both I and II are correct
- (2) Both I & II are incorrect
- (3) I is correct, II is incorrect
- (4) I is incorrect, II is correct

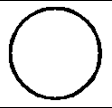
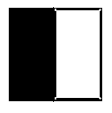
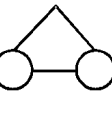
173. What will be the percentage of inheritance of a x-linked disorder from diseased father to his son ?

- (1) 0%
- (2) 25%
- (3) 50%
- (4) 100%

174. Which of the following structure was discovered by Henking ?

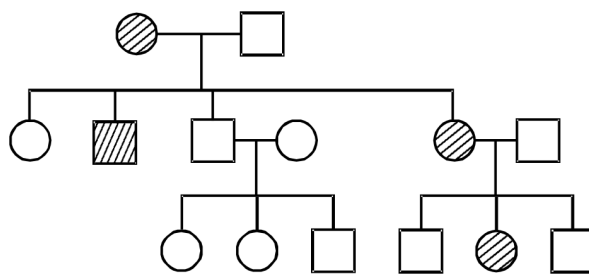
- (1) y-body
- (2) Bar body
- (3) x-body
- (4) Nu-body

175. Match the following.

(i)		(A)	Monozygotic twins
(ii)		(B)	Carrier for sex linked recessive disease
(iii)		(C)	Carrier for autosomal recessive disease
(iv)		(D)	Normal female

- (1) (i) – A, (ii) – B, (iii) – C, (iv) – D
- (2) (i) – D, (ii) – C, (iii) – A, (iv) – B
- (3) (i) – C, (ii) – D, (iii) – B, (iv) – A
- (4) (i) – B, (ii) – C, (iii) – D, (iv) – A

176. Given pedigree show inheritance of autosomal dominant disorder (myotonic dystrophy) then what will be genotype of shaded symbol :-



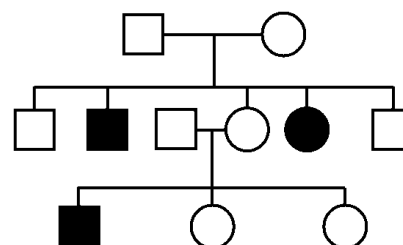
- (1) AA
- (2) Aa
- (3) AA or Aa
- (4) aa

177. According to Hardy-Weinberg law, the frequency of organisms having dominant character will be –
 (1) p^2 (2) q^2
 (3) $p^2 + 2pq$ (4) $p^2 + 2pq + q^2$
178. If the frequency of dominant allele in a population is 0.7. The frequency of recessive phenotype would be :-
 (1) 0.30 (2) 0.09 (3) 0.49 (4) 0.91
179. Which of the following is not a component of innate immunity ?
 (1) Interferons (2) Phagocytes
 (3) NK-cells (4) Antibodies
180. Which antibody is present in pentamer form :-
 (1) IgG (2) IgM (3) IgE (4) IgD
181. Which type of immunity is responsible for graft rejection :-
 (1) Cell mediated immunity
 (2) Antibody mediated immunity
 (3) Humoral immunity
 (4) Innate immunity
182. Which of the following is not used to reduce symptoms of allergy ?
 (1) Adrenaline (2) Cortisol
 (3) Histamine (4) Cetirizine
183. **Assertion (A) :-** Macrophages act as HIV-factory.
Reason (R) :- T_H cells die but macrophages survive the process of HIV replication.
 (1) Both A and R are true but R is not correct explanation of A.
 (2) Both A & R are false
 (3) A is true but R is false.
 (4) Both A & R are true and R is correct explanation of A.
184. Inflammation of Synovial membrane due to accumulation of antigen – antibody complexes produces
 (1) rheumatoid arthritis
 (2) gout
 (3) dislocation
 (4) carcinoma
185. Severe combined immuno deficiency is due to deficiency of enzyme :-
 (1) Adenosine deaminase
 (2) Tyrosinase
 (3) Phenylalanine hydroxylase
 (4) Histamine

SECTION-B (BIOLOGY-II)

186. Read the following statements carefully and choose the correct option :-
 (a) Phenotype is the external manifestation of gene product
 (b) Individuals with similar phenotypes may not belong to same genotype
 (c) Individuals with different phenotypes always have similar genotypes
 (d) Phenotype is not influenced by environment
 (e) Human height is influenced by environment.
 (1) b, c & d are correct (2) a, b & e are correct
 (3) c & d are correct (4) a & b are correct
187. The four types of phenotypes are obtained in F_2 of dihybrid cross in garden pea due to _____ but yellow round seeded plants are maximum in number due to _____ :-
 (1) Independent assortment, segregation
 (2) Independent assortment, dominance
 (3) Segregation, independent assortment
 (4) Independent assortment, linkage

188. In a dihybrid cross 1280 plants were produced in f_2 generation. If gene for first character shows dominance and gene for second character shows incomplete dominance then how many of these plants are showing dominant trait for first character and intermediate trait for second character ?
 (1) 240 (2) 480 (3) 620 (4) 320
189. If the genotypes of husband and wife are $I^A I^B$ and $I^A I^O$: Among the blood types of their childrens, how many different genotypes and phenotypes are possible ?
 (1) 2 genotypes, 3 phenotypes
 (2) 3 genotypes, 4 phenotypes
 (3) 4 genotypes, 4 phenotypes
 (4) 4 genotypes, 3 phenotypes
190. Chromosomal theory of inheritance is the united form of :-
 (1) Mendelian principles and cell division
 (2) Knowledge of chromosomal segregation and Boveri principles
 (3) Knowledge of chromosomal segregation and Mendelian principles
 (4) Knowledge of chromosomal assortment and chromosomal segregation
191. Which of the following characters of *Drosophila* is suitable for genetical studies?
 (a) They could be grown on simple synthetic medium in laboratory
 (b) They complete their life cycle in about 2-weeks
 (c) Single mating produces few number of progeny flies.
 (d) They have many types of hereditary variations that can be seen with low power microscope.
 (e) Male flies are larger than female flies
 Select the correct option:-
 (1) a, b, d (2) a, b, c, d, e
 (3) d and e (4) c and e
192. The cross over frequencies between the genes A & B, A & C, and B & C is 6%, 15% & 21% respectively. What is the sequence of genes on chromosome :-
 (1) A, B, C
 (2) B, A, C
 (3) A, C, B
 (4) Either B, A, C or C, A, B,
193. Which of the following statement is true for colourblindness :-
 (1) It is due to mutation in certain genes present in y-chromosome
 (2) It occurs in about 8% of female and about 0.4 % of male
 (3) The son of a woman who carries the gene has 50% chance of being colourblind
 (4) It is an autosomal linked disorder
194. If somatic cells of human male contain single barrbody, the genetic composition of the person would be
 (1) XYY
 (2) XXY
 (3) XO
 (4) XXXY
195. The given pedigree chart shows the inheritance of :-



- (1) Autosomal dominant - Myotonic dystrophy.
 (2) X-linked dominant - Pseudoricketes
 (3) Autosomal recessive - Sickle cell anaemia
 (4) X-linked recessive - Haemophilia.

- 196.** Which scientist disapproved the "Good humor hypothesis" ?

 - (1) Hippocrates
 - (2) Indian Ayurvedic System of Medicine
 - (3) William Harvey
 - (4) (1) and (2)

197. Which of the following set of cells does not participate in acquired active immunity ?

 - (1) NK-cells, TH cells
 - (2) TH cells, B-Lymphocytes
 - (3) NK-cells, Neutrophils
 - (4) Neutrophils, T-killer cells

198. Maximum concentration of histamine found in case of ?

 - (1) Anaphylactic shock
 - (2) Hay fever
 - (3) Bronchial asthma
 - (4) Pneumonia

199. Tumor cells have been shown to avoid detection and destruction by immune system. Therefore the patient are given certain substances called biological response modifiers (BRM) which activates their immune system and helps in destroying the tumor. Identify the BRM :-

 - (1) Alpha-interferon
 - (2) Alpha lactalbumin
 - (3) Alpha-1 antitrypsin
 - (4) Gamma globulin

200. How many of following are not autosomal recessive gene mutation disorder ?

 - (a) Phenyl Ketoneuria
 - (b) Alkaptonuria
 - (c) Tay-sach disease
 - (d) Albinism
 - (e) Glucose-6 phosphate dehydrogenase deficiency disease
 - (f) Duchenne Muscular Dystrophy.

(1) One (2) Three
(3) Four (4) Two

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Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005
Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.ac.in | Website : www.allen.ac.in